Trace a Web/API Request Using DevTools, DNS, and curl

Objective: To demonstrate how one API request can be traced from DNS resolution to browser inspection and command-line response validation.
Tools required: Google Chrome with DevTools, Terminal, nslookup, curl, Internet access

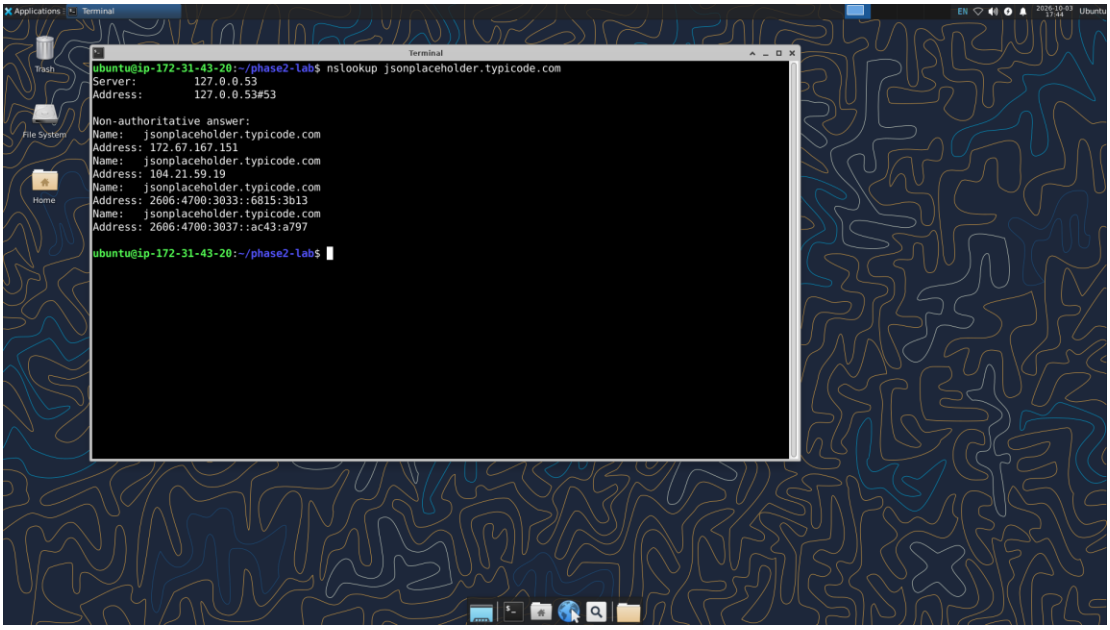
Steps to be followed:

1. Resolve the API hostname using DNS
2. Open the API endpoint in Chrome
3. Capture the request in Chrome DevTools
4. Inspect the request method, URL, and status
5. Inspect request and response headers
6. Inspect the JSON response body
7. Send the same request using curl and verify the result

Step 1: Resolve the API hostname using DNS

Use nslookup to resolve the public API hostname. The returned IP address may change over time.

```
nslookup jsonplaceholder.typicode.com
```

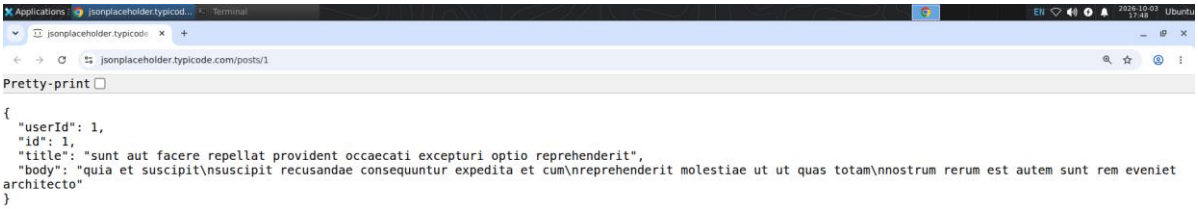


DNS lookup for jsonplaceholder.typicode.com

Step 2: Open the API endpoint in Chrome

Open the following endpoint and confirm that the browser displays a JSON response.

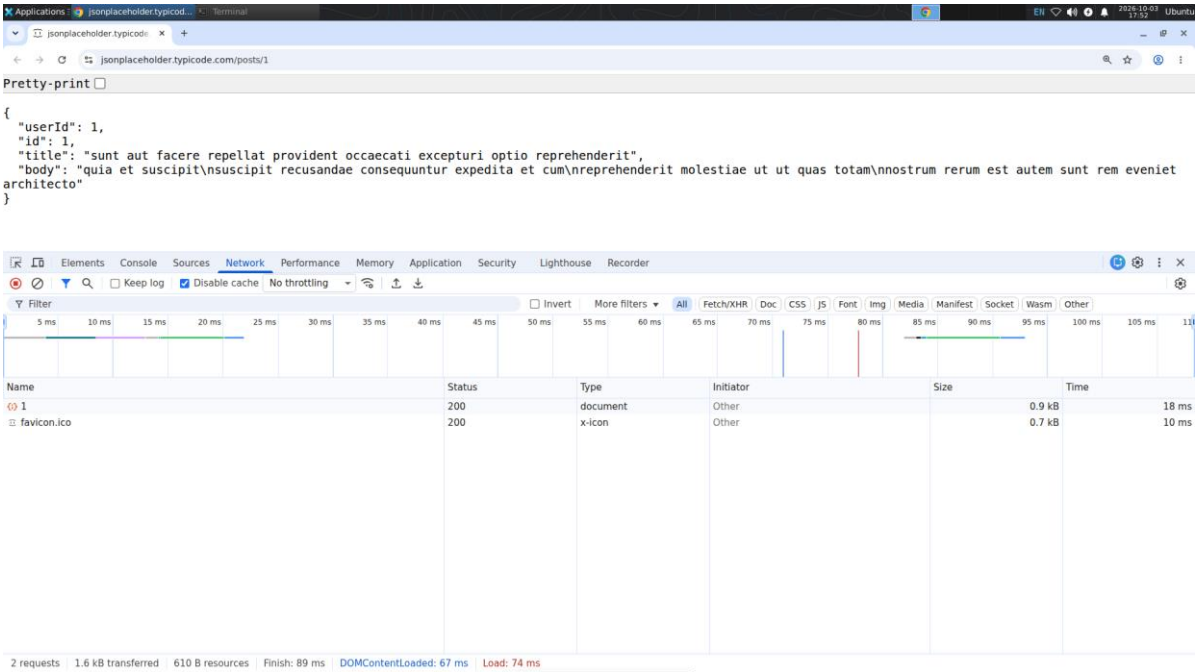
```
https://jsonplaceholder.typicode.com/posts/1
```



API endpoint opened in Chrome

Step 3: Capture the request in Chrome DevTools

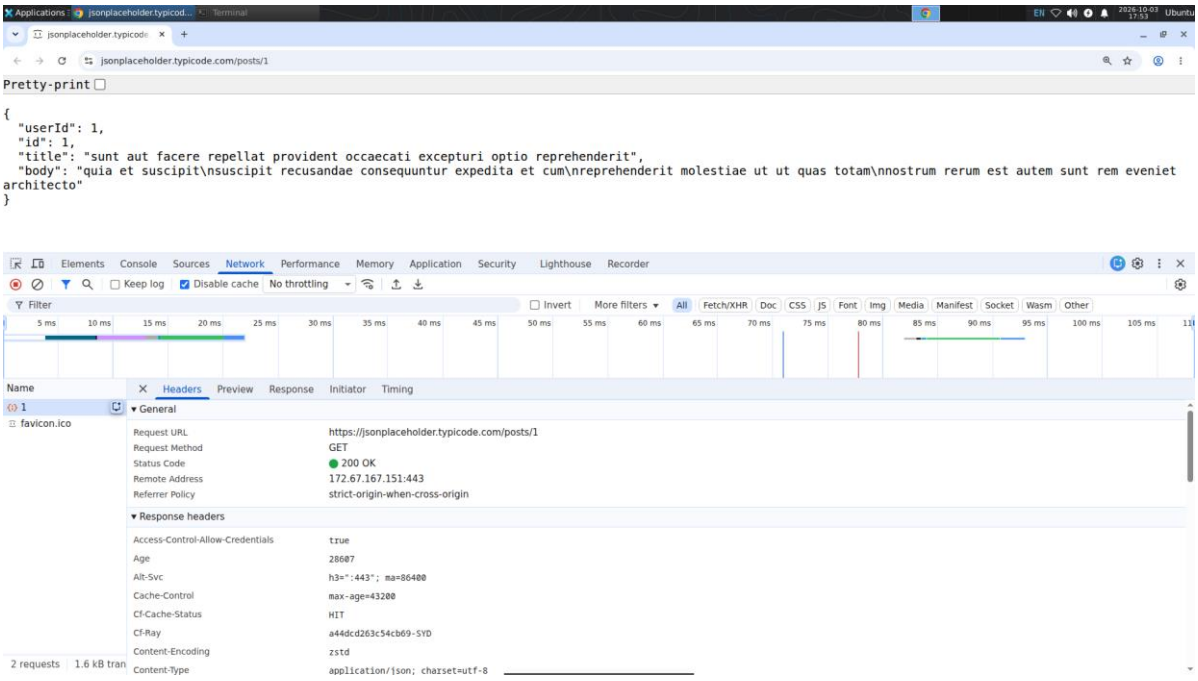
Open Developer Tools, select the Network tab, and reload the page so the API request is captured.



API request captured in the Chrome DevTools Network panel

Step 4: Inspect the request method, URL, and status

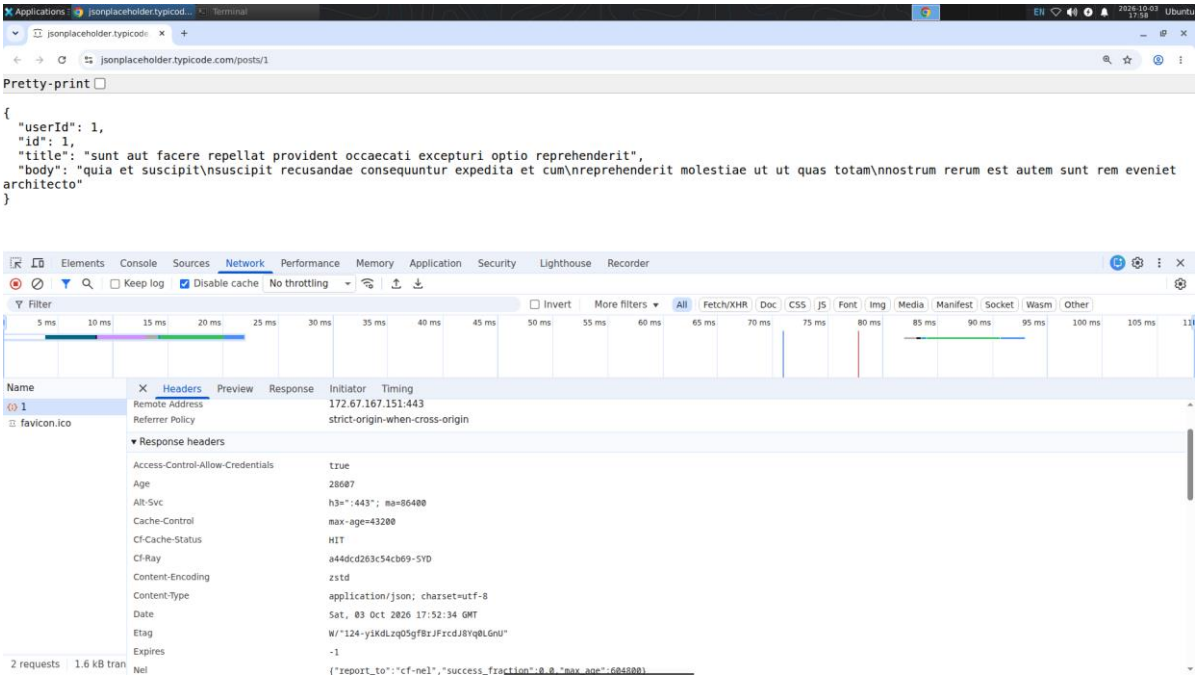
Select the /posts/1 request. Verify that the request uses GET and returns a successful HTTP status.



Request details showing URL, GET method, and successful status

Step 5: Inspect request and response headers

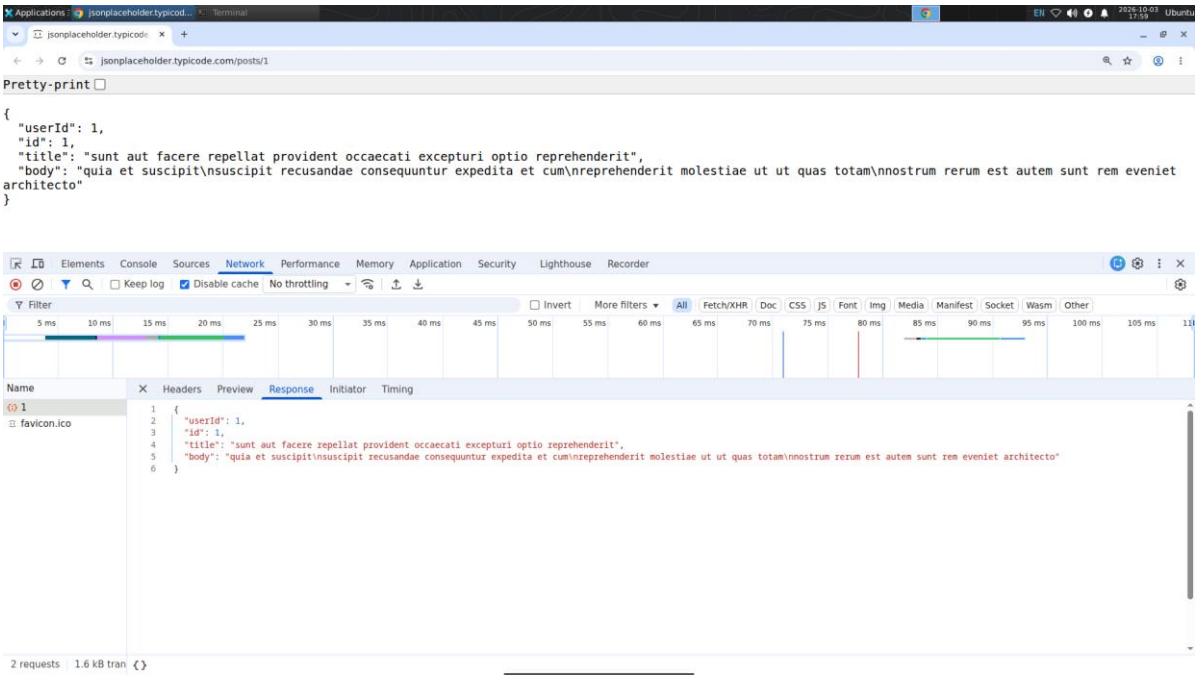
Review the Headers area and identify the request URL, method, status, and response content type.



Request and response headers in Chrome DevTools

Step 6: Inspect the JSON response body

Open the Response or Preview tab and connect the GET request to the returned JSON data.

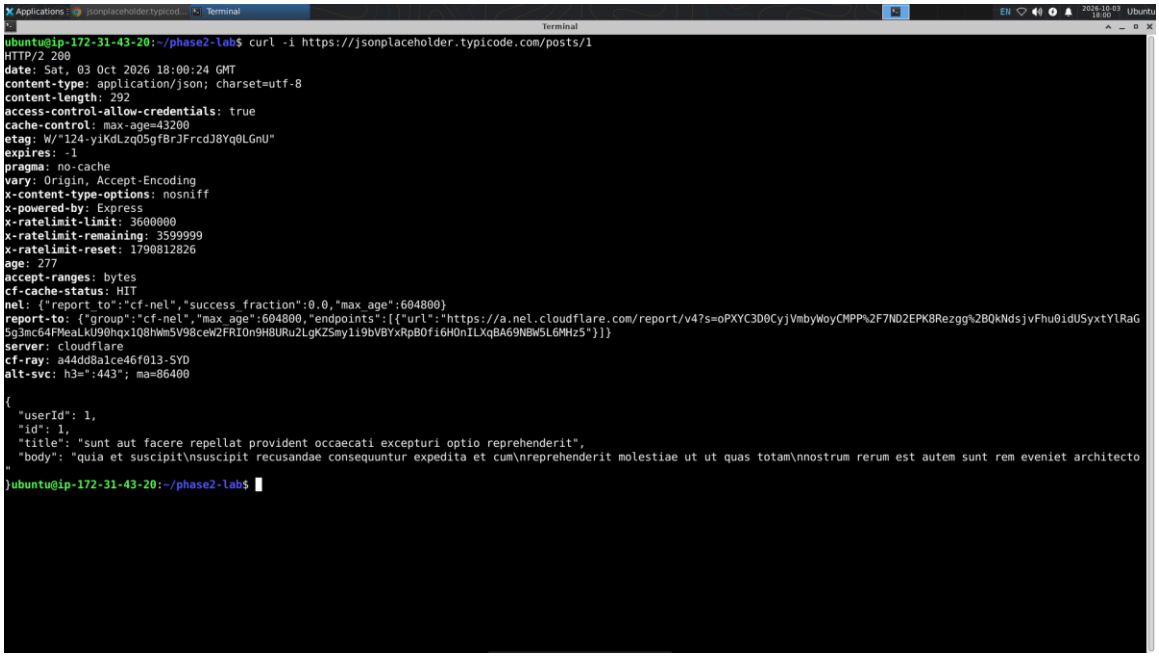


JSON response body returned by the API

Step 7: Send the same request using curl

Use curl with the -i option to display the HTTP response headers and JSON body together.

```
curl -i https://jsonplaceholder.typicode.com/posts/1
```



Successful API response using curl

Step 8: Verify the request journey

Compare the browser, DevTools, DNS, and curl evidence. The same API interaction can be observed from different viewpoints.

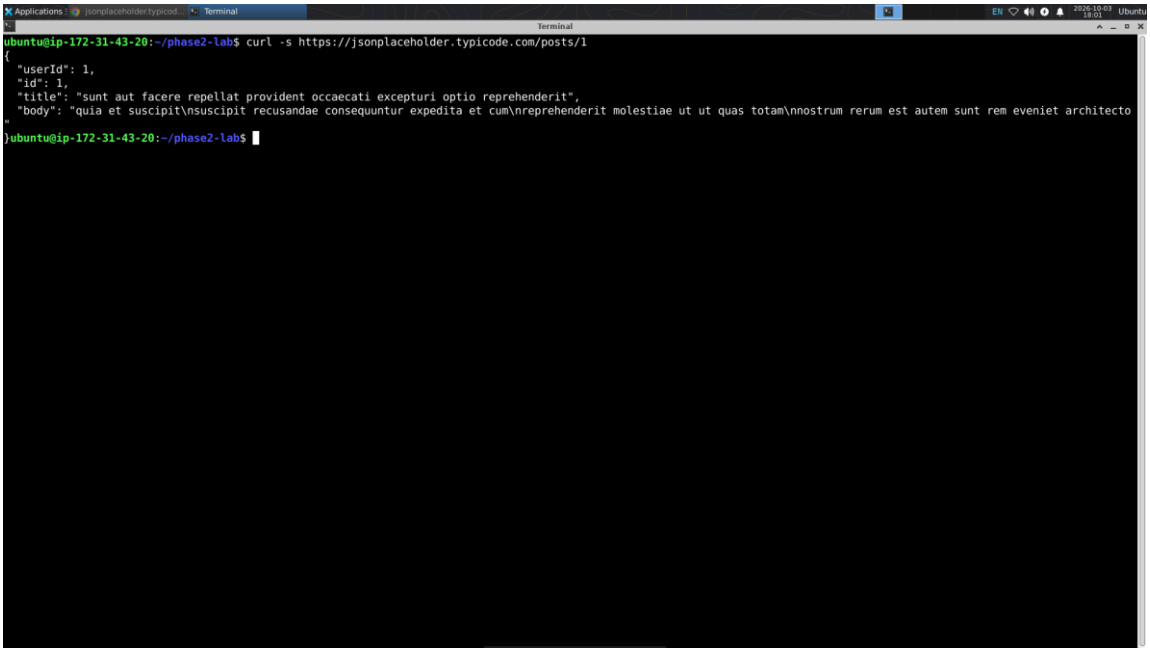
Verification

- The API hostname resolves successfully using DNS.
- The API endpoint opens in Chrome and returns JSON.
- Chrome DevTools captures the /posts/1 request.
- The request method is GET and the response status is successful.
- Request/response headers and the JSON response body can be inspected.
- curl sends the same request and displays the HTTP response and JSON body.

Activity complete when all verification points are satisfied.

Optional reference: JSON-only curl output

```
curl -s https://jsonplaceholder.typicode.com/posts/1
```



Optional clean JSON-only response using curl